

# Math 74: Algebraic Topology

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## Problem 1.

- (a) Let  $p : (E, e_0) \rightarrow (B, b_0)$  be a covering map. Suppose that  $E$  is path-connected. Then show that there is a surjection,

$$\phi : \pi_1(B, b_0) \rightarrow p^{-1}(b_0)$$

- (b) Moreover, if  $E$  is simply connected, then  $\phi$  is a bijection.

**Hint:** Try to modify the computation of  $\pi_1(S^1, (1,0))$  for the first and second part.

*Solution.*

- (a) For  $[\gamma] \in \pi_1(B, b_0)$ , let  $\gamma' : I \rightarrow E$  be the unique lift of  $\gamma$  starting at  $e_0$ . Define

$$\phi : \pi_1(B, b_0) \rightarrow p^{-1}(b_0), \quad \phi([\gamma]) = \gamma'(1).$$

Note  $\gamma'(1) \in p^{-1}(b_0)$  since  $p(\gamma'(1)) = \gamma(1) = b_0$ , so  $\phi$  is well-defined.

To see surjectivity, let  $e_\alpha \in p^{-1}(b_0)$ . Since  $E$  is path-connected, there exists a path  $\sigma_\alpha : I \rightarrow E$  such that  $\sigma_\alpha(0) = e_0$  and  $\sigma_\alpha(1) = e_\alpha$ .

Then  $\gamma = p \circ \sigma_\alpha$  is a loop in  $B$  based at  $b_0$ , since

$$\gamma(0) = (p \circ \sigma_\alpha)(0) = p(e_0) = b_0 \quad \text{and} \quad \gamma(1) = (p \circ \sigma_\alpha)(1) = p(e_\alpha) = b_0$$

By uniqueness of lifts,  $\sigma_\alpha$  is the unique lift of  $\gamma$  starting at  $e_0$ , so  $\phi([\gamma]) = \sigma_\alpha(1) = e_\alpha$ . We can similarly define a unique lift for each  $e_\alpha$  and thus  $\phi$  is surjective.

- (b) By part (a) we already know that  $\phi$  is surjective. To see that  $\phi$  is injective, let  $[f], [g] \in \pi_1(B, b_0)$  such that  $\phi([f]) = \phi([g]) = e_\alpha$  for some  $e_\alpha \in p^{-1}(b_0)$ , and let  $f', g'$  be the lifts of  $f, g$  starting at  $e_0$ .

Then  $f'(0) = g'(0) = e_0$  and  $f'(1) = g'(1) = e_a$ . Since  $E$  is simply connected,  $\pi_1(E, e_0) \cong \{e\}$ , i.e. any two paths with the same initial and final points in  $E$  are path-homotopic. Thus,  $[f'] = [g']$ , let  $F'$  be the homotopy between  $f'$  and  $g'$ . Then we have

$$F'(s, 0) = f'(s) \quad \text{and} \quad F'(s, 1) = g'(s).$$

Applying  $p$  to both sides,

$$(p \circ F')(s, 0) = p(f'(s)) = f(s) \implies p(f') = f$$

$$(p \circ F')(s, 1) = p(g'(s)) = g(s) \implies p(g') = g$$

Therefore  $p \circ F' : I \times I \rightarrow B$  is a homotopy between  $f$  and  $g$ . Hence,  $[f] = [g]$ ,  $\phi$  is injective, and thus bijective.

**Problem 2.** Let  $p : E \rightarrow B$  be a covering map, with  $B$  connected. Then if  $p^{-1}(b_0)$  has  $k$ -many elements for some finite number  $k$ ,  $p^{-1}(b)$  also has  $k$ -many elements for every  $b \in B$ .

*Solution.*

Assume  $p^{-1}(b_0)$  has  $k$  elements, and let  $U$  be an evenly covered neighborhood of  $b_0$ , so that

$$p^{-1}(U) = \bigsqcup_{i=1}^k V_i$$

where each  $V_i$  maps homeomorphically onto  $U$ . For any  $b \in U$ , the preimage  $p^{-1}(b)$  contains exactly one point from each  $V_i$ , hence has exactly  $k$  elements.

Now let  $S = \{b \in B : |p^{-1}(b)| = k\}$  and  $T = \{b \in B : |p^{-1}(b)| \neq k\}$ , so  $B = S \sqcup T$ . We know that every point in  $S$  has an evenly covered neighborhood contained entirely in  $S$ , so  $S$  is open. Similarly, every point in  $T$  has an evenly covered neighborhood contained entirely in  $T$ , so  $T$  is open. Since  $B$  is connected and  $S$  is nonempty (as  $b_0 \in S$ ), we must have  $T = \emptyset$ , i.e.  $S = B$ . Therefore  $|p^{-1}(b)| = k$  for all  $b \in B$ .

**Problem 3.** If  $p : E \rightarrow B$  is a covering map with  $E$  path-connected and  $B$  simply-connected. Then  $p$  is a homeomorphism.

*Solution.*

Let  $p : E \rightarrow B$  be a covering map, where  $E$  is path-connected and  $B$  is simply-connected.

- **Surjectivity:** Covering maps are surjective by definition.
- **Injectivity:** Since  $B$  is simply-connected,  $\pi_1(B, b_0) = \{e\}$  is trivial. By Problem 1(a), the map

$$\phi : \pi_1(B, b_0) \rightarrow p^{-1}(b_0), \quad [\gamma] \mapsto \tilde{\gamma}(1)$$

is surjective. Since the domain has exactly one element,  $p^{-1}(b_0)$  has exactly one element. By Problem 2, this means that we have a 1-fold cover for  $B$ , i.e.,  $|p^{-1}(b)| = 1$  for all  $b \in B$ . Thus  $p$  is injective.

- **Continuity of  $p^{-1}$ :** Let  $V \subseteq E$  be an open set. For each  $e \in V$ , let  $U_e$  be an evenly covered neighborhood of  $p(e)$  in  $B$ . Since  $p$  is a 1-fold covering,  $p^{-1}(U_e) = S_e$  is the unique disjoint set covering  $U_e$ , and

$$p|_{S_e} : S_e \rightarrow U_e$$

is a homeomorphism. Then  $p(V \cap S_e) = p|_{S_e}(V \cap S_e)$  is open in  $U_e \subseteq B$ , and contains  $p(e)$ . Then,

$$p(V) = \bigcup_{e \in V} p(V \cap S_e)$$

is a union of open sets, hence open. Therefore  $p^{-1}$  is continuous.

Hence  $p$  is a continuous bijective map with a continuous inverse and thus a homeomorphism.

**Problem 4.** Show that if  $p : E \rightarrow B$  and  $p' : E' \rightarrow B'$  are covering maps, then so is  $p \times p' : E \times E' \rightarrow B \times B'$ , where  $p \times p'$  is defined in an obvious manner.

*Solution.*

We define  $p \times p' : E \times E' \rightarrow B \times B'$  by

$$(p \times p')(e, e') = (p(e), p'(e')).$$

This is continuous since each component is continuous.

**Surjectivity.** Given any  $(b, b') \in B \times B'$ , since  $p$  and  $p'$  are surjective there exist  $e \in E$  and  $e' \in E'$  with  $p(e) = b$  and  $p'(e') = b'$ , so  $(p \times p')(e, e') = (b, b')$ .

**Evenly covered.** Let  $(b, b') \in B \times B'$ . Since  $p$  is a covering map, there exists an open neighborhood  $U$  of  $b$  in  $B$  that is evenly covered, with

$$p^{-1}(U) = \bigsqcup_{\alpha} V_{\alpha},$$

where each  $V_{\alpha}$  is open in  $E$  and  $p|_{V_{\alpha}} : V_{\alpha} \rightarrow U$  is a homeomorphism. Since  $p'$  is a covering map, there exists an open neighborhood  $U'$  of  $b'$  in  $B'$  that is evenly covered, with

$$p'^{-1}(U') = \bigsqcup_{\beta} V'_{\beta},$$

where each  $V'_{\beta}$  is open in  $E'$  and  $p'|_{V'_{\beta}} : V'_{\beta} \rightarrow U'$  is a homeomorphism. Then  $U \times U'$  is an open neighborhood of  $(b, b')$  in  $B \times B'$ , and

$$(p \times p')^{-1}(U \times U') = p^{-1}(U) \times p'^{-1}(U') = \left( \bigsqcup_{\alpha} V_{\alpha} \right) \times \left( \bigsqcup_{\beta} V'_{\beta} \right) = \bigsqcup_{\alpha, \beta} (V_{\alpha} \times V'_{\beta}).$$

Each  $V_{\alpha} \times V'_{\beta}$  is open in  $E \times E'$ , and the restriction

$$(p \times p')|_{U_{\alpha} \times U'_{\beta}} = p|_{U_{\alpha}} \times p'|_{U'_{\beta}} : U_{\alpha} \times U'_{\beta} \rightarrow U \times U'$$

is a product of homeomorphisms, hence a homeomorphism. Hence  $U \times U'$  is evenly covered, and  $p \times p'$  is a covering map.

**Problem 5.** Show that if a subspace  $A$  is a deformation retraction of a space  $X$ , and another subspace  $B$  is a deformation retraction of  $A$ , then  $B$  is a deformation retraction of  $X$ .

*Solution.*

Let  $H : X \times I \rightarrow X$  be the deformation retraction to  $A$ , so that  $H(x, 0) = x$ ,  $H(x, 1) \in A$ , and  $H(a, t) = a$  for all  $x \in X$ ,  $a \in A$ ,  $t \in I$ . Let  $G : A \times I \rightarrow A$  be the deformation retraction to  $B$ , so that  $G(a, 0) = a$ ,  $G(a, 1) \in B$ , and  $G(b, t) = b$  for all  $a \in A$ ,  $b \in B$ ,  $t \in I$ .

Note that since  $H(x, 1) \in A$  for all  $x \in X$ , the expression  $G(H(x, 1), t)$  is well-defined for all  $t \in I$ . Now, we define the deformation retraction  $P : X \times I \rightarrow X$  (from  $X$  to  $B$ ) by,

$$P(x, t) = \begin{cases} H(x, 2t) & t \in [0, \frac{1}{2}] \\ G(H(x, 1), 2t - 1) & t \in [\frac{1}{2}, 1] \end{cases}$$

At  $t = \frac{1}{2}$  we have  $H(x, 1)$  and  $G(H(x, 1), 0) = H(x, 1)$  respectively, so  $P$  is well-defined. Moreover, since both  $H$  and  $G$  are continuous,  $P$  is continuous by the pasting lemma.

We verify the three conditions for a deformation retraction to  $B$ :

- **Starts as identity:**  $P(x, 0) = H(x, 0) = x$ .
- **Ends in  $B$ :**  $P(x, 1) = G(H(x, 1), 1) \in B$ .
- **$B$  is fixed throughout:** Let  $b \in B \subseteq A$ . For  $t \in [0, \frac{1}{2}]$ ,  $P(b, t) = H(b, 2t) = b$  since  $b \in A$ . For  $t \in [\frac{1}{2}, 1]$ ,  $P(b, t) = G(H(b, 1), 2t - 1) = G(b, 2t - 1) = b$  since  $b \in B$ .

Hence,  $P$  is a deformation retraction from  $X$  to  $B$ .

**Problem 6.**

- (a) Show that if a space retracts onto a subspace  $A$ , then the homomorphism on the fundamental group induced by the inclusion  $\iota : A \hookrightarrow X$  is injective.
- (b) Also show that if  $A$  is a deformation retract of  $X$ , then  $\iota_*$  is an isomorphism.

*Solution.*

- (a) Let  $r$  be the map that retracts  $X$  to  $A$ , and  $r(x_0) = a_0$  in  $A$ . Note that the composition  $r \circ \iota$  is the identity on  $A$ . Now consider the homomorphism  $(r \circ \iota)_*$ . We have,

$$(r \circ \iota)_* = r_* \circ \iota_* = (\text{id}_A)_* = \text{id}_{\pi_1(A, a_0)}$$

Let  $\iota_*([f]) = \iota_*([g])$ . Then,

$$\begin{aligned} r_*(\iota_*([f])) &= r_*(\iota_*([g])) \\ \implies (\text{id}_A)_*([f]) &= (\text{id}_A)_*([g]) \\ \implies [f] &= [g] \end{aligned}$$

Hence  $\iota_*$  is injective.

- (b) Let  $H$  be the deformation retraction of  $X$  to  $A$ , and  $H(x_0, 1) = a_0$  in  $A$ . From part (a) we know that  $\iota_*$  is injective. So we only need to show that  $\iota_*$  is surjective.

We define the continuous map  $r = H(x, 1)$ . Consider the map  $\iota \circ r$  which maps  $X \rightarrow A \rightarrow X$ . This map is homotopic to the identity map by  $H$  (by definition),

$$\iota \circ r \simeq \text{id}_X$$

Then  $(\iota \circ r)_* = \iota_* \circ r_* \simeq (\text{id}_X)_* = \text{id}_{\pi_1(X, x_0)}$  (Lemma 58.1). Since  $\text{im}(r_*) = \pi_1(A, a_0)$ ,  $\iota_*$  is surjective. Thus, we have an isomorphism  $\pi_1(A, a_0) \cong \pi_1(X, x_0)$ .

**Problem 7.** Show that if a space  $X$  deformation retracts to a point  $x \in X$ , then for each neighborhood  $U$  of  $x$  in  $X$  there exists a neighborhood  $V \subset U$  of  $x$  such that the inclusion map  $V \hookrightarrow U$  is nullhomotopic (nullhomotopic means homotopic to a constant map).

*Solution.*

Let  $H : X \times I \rightarrow X$  be the deformation retraction from  $X$  to  $\{x\}$  such that  $\forall x' \in X$ ,

$$H(x', 0) = x', \quad H(x', 1) = x, \text{ and } H(x, t) = x$$

and let  $U \subseteq \text{im}(H) = H$  be an open set containing  $x$ . Note that the preimage  $H^{-1}(U)$  is open in  $X \times I$  and contains  $\{x\} \times I$  (since  $x \in U$ ).

We claim there exists an open neighborhood  $V$  of  $x$  in  $X$  such that  $V \times I \subseteq H^{-1}(U)$ .

For each  $t \in I$ , continuity of  $H$  at  $(x, t)$  gives a open set  $V_t \times W_t$  containing  $(x, t)$  with  $H(V_t \times W_t) \subseteq U$ . The sets  $\{W_t\}_{t \in I}$  form an open cover of  $I$ . Since  $I$  is compact, there is a finite subcover  $W_{t_1}, \dots, W_{t_n}$ . Let  $V = V_{t_1} \cap \dots \cap V_{t_n}$  (finite intersection of open sets containing  $x$  is an open set containing  $x$ ). Note that for any  $(y, t) \in V \times I$ , we have  $t \in W_{t_k}$  for some  $k$  and  $y \in V \subseteq V_{t_k}$ , so  $H(y, t) \in U$ .

Hence there exists some open set  $V \times I \subseteq H^{-1}(U)$  and the restriction  $H|_{V \times I} : V \times I \rightarrow U$  is a homotopy from the inclusion  $\iota : V \hookrightarrow U$  (at  $t = 0$ ) to the constant map at  $x$  (at  $t = 1$ ) so  $\iota \simeq e_x$ .

**Problem 8.**

- (a) Let  $X$  be the subspace of  $\mathbb{R}^2$  consisting of the horizontal segment  $[0, 1] \times 0$  together with all the vertical segments  $r \times [0, 1 - r]$  for  $r$  a rational number in  $[0, 1]$ . Show that  $X$  deformation retracts to any point in the segment  $[0, 1] \times 0$ , but not to any other point. [See the preceding problem.]
- (b) Let  $Y$  be the subspace of  $\mathbb{R}^2$  that is the union of an infinite number of copies of  $X$  arranged as in the figure below. Show that  $Y$  is contractible but does not deformation retract onto any point.



*Solution.*

- (a) Let  $a = (x_0, 0)$  be an arbitrary point in  $[0, 1] \times \{0\}$ . It suffices to show that  $X$  deformation retracts to the subspace  $B = [0, 1] \times \{0\}$ , since  $B$  is convex and hence deformation retracts to any of its points via the straight-line homotopy, and by Problem 5 the composition of deformation retracts is a deformation retract. We define  $H : X \times I \rightarrow X$  as follows:

$$H((x, y), t) = (1 - t)(x, y) + t(x, 0)$$

Note that

$$H((x, y), 0) = (x, y) \in X, \quad H((x, y), 1) = (x, 0) \in B$$

And

$$H((x, 0), t) = (1 - t)(x, 0) + t(x, 0) = (x, 0) \in B$$

So  $H$  is a deformation retraction from  $X$  to  $B$ . Since  $B \cong I$  is convex, it deformation retracts to  $a$  via

$$G((x, 0), t) = ((1 - t)x + tx_0, 0).$$

By Problem 5,  $X$  deformation retracts to  $\{a\}$ .

We show that  $X$  does not deformation retract to any point  $a = (r, h)$ ,  $h > 0$  by showing the contrapositive of the condition in Problem 7. That is there exists some neighborhood  $U \subseteq X$  of  $x$  such that for all neighborhoods  $V \subset U$  of  $x$ , the inclusion map is not nullhomotopic then  $X$  does not deformation retract to  $a \in X$ .

Define  $0 < \varepsilon < \min(h, 1 - r)$  and let  $U$  be the  $\varepsilon$ -ball centered at  $a$  in  $X$ ,

$$U = B_\varepsilon(a) \cap X$$

That is,  $U$  does not intersect the  $x$ -axis and hence is composed of infinitely many disjoint line segments (one for each rational  $x$  with  $|x - r| < \varepsilon$ .)

$$U = \bigsqcup_{\substack{r \in \mathbb{Q} \\ |x-r| < \varepsilon}} (\{r\} \times (h - \varepsilon, h + \varepsilon)) \cap X.$$

Now let  $V \subseteq U$  be any sub-neighborhood of  $a$ . Then  $V$  is also composed of infinitely many disjoint line segments.

We claim the inclusion  $\iota : V \hookrightarrow U$  is not nullhomotopic. If it were, there would exist a homotopy

$$H : V \times I \rightarrow U$$

with  $H(v, 0) = v$  and  $H(v, 1) = u_0$  for some fixed  $u_0 \in U$ .

In particular, for each  $v \in V$ , there must be some path connecting  $v$  to  $u_0$  inside  $U$ . But  $v$  and  $u_0$  may lie in different path components of  $U$ , making this impossible. Hence there exists a  $U$  such that for all  $V \subset U$  the inclusion map is not nullhomotopic and thus  $X$  does not deformation retract to  $a \in X$ .

(b) **TODO: contractible**

We show that  $Y$  does not deformation retract to any point. By part (a), within each copy of  $X$ , the only candidates for a deformation retract are points on the base  $x$ -axis. The only points that qualify are those that are on the corner of two adjacent copies of  $X$ , since they lie on the base of both  $X$ s. So it suffices to show that  $Y$  does not deformation retract to any such  $a$ .

For the sake of contradiction assume it retracts to a point  $a = (x_0, y_0)$ . WLOG let  $a = (0, 1)$ . Define  $\varepsilon > 0$  and let  $U$  be the  $\varepsilon$ -ball centered at  $a$  in  $Y$ ,

$$U = B_\varepsilon(a) \cap Y$$

That is,  $U$  is composed of infinitely many disjoint line segments. Now let  $V \subseteq U$  be any sub-neighborhood of  $a$ . Then  $V$  contains an open ball  $B_\delta(a)$  for some  $\delta > 0$ . Consider the sequence of points

$$a_n = \left( \frac{1}{n}, 1 - \frac{1}{n} \right).$$

Each  $a_n$  lies in  $Y$  since  $1/n$  is rational, and each  $a_n$  is the tip of a distinct vertical segment, hence lies in a distinct path component of  $U$ . Since  $\lim_{n \rightarrow \infty} a_n \rightarrow a$ , the tail of the sequence eventually lies in  $B_\delta(a) \subseteq V$ .

Suppose  $\iota : V \hookrightarrow U$  were nullhomotopic. Then there would be a homotopy  $H : V \times I \rightarrow U$  with  $H(v, 0) = v$  and  $H(v, 1) = u_0$  for some fixed  $u_0 \in U$  and all  $v \in V$ . In particular, for each  $v \in V$ , there must be some path connects  $v$  to  $u_0$  inside  $U$ . But for large enough  $n$ ,  $a_n \in V$  lies in a different path component of  $U$  than  $u_0$ , so no such path can exist. This is a contradiction, so no such  $V$  exists satisfying the conclusion of Problem 7. Therefore  $Y$  does not deformation retract to  $a$ .

**Problem 9.** Define  $f : S^1 \times I \rightarrow S^1 \times I$  by  $f(\theta, s) = (\theta + 2\pi s, s)$ , so  $f$  restricts to the identity on the two boundary circles of  $S^1 \times I$ . Show that  $f$  is homotopic to the identity by a homotopy  $f_t$  that is stationary on one of the boundary circles, but not by any homotopy  $f_t$  that is stationary on both boundary circles. [Consider what  $f$  does to the path  $s \mapsto (\theta_0, s)$  for fixed  $\theta_0 \in S^1$ ].

*Solution.*

To show that  $f$  is homotopic to the identity, we define  $f_t : S^1 \times I \rightarrow S^1 \times I$  as follows:

$$f_t(\theta, s) = (\theta + 2\pi s(1 - t), s)$$

We check,

$$f_0(\theta, s) = (\theta + 2\pi s, s) = f, \quad f_1(\theta, s) = (\theta, s) = \text{id}, \quad f_t(\theta, 0) = (\theta, 0) = \text{id} \mid_{S^1 \times \{0\}}$$

Hence,  $f$  is homotopic to the identity by a homotopy  $f_t$  that is stationary on one of the boundary circles.

Now suppose for contradiction that there exists a homotopy  $f_t : S^1 \times I \rightarrow S^1 \times I$  between  $f$  and the identity that is stationary on both boundary circles. Let  $\theta_0$  be some point in  $S^1$  and consider the vertical path

$$\gamma : I \rightarrow S^1 \times I, \quad \gamma(s) = (\theta_0, s).$$

Since  $f_t$  is stationary on both boundary circles, we have  $f_t(\theta_0, 0) = (\theta_0, 0)$  and  $f_t(\theta_0, 1) = (\theta_0, 1)$  for all  $t$ . Therefore  $f_t \circ \gamma$  is a homotopy between  $\gamma$  and  $f \circ \gamma$ , where

$$(f \circ \gamma)(s) = (\theta_0 + 2\pi s, s).$$

Now let  $\pi : S^1 \times I \rightarrow S^1$  be the projection  $\pi(\theta, s) = \theta$ . Applying  $\pi$ , we get a homotopy in  $S^1$  between

$$\pi \circ \gamma \equiv \theta_0 \quad (\text{the constant path})$$

and

$$\pi \circ f \circ \gamma : s \mapsto \theta_0 + 2\pi s \quad (\text{a loop winding once around } S^1).$$

But these two loops represent different elements of  $\pi_1(S^1, \theta_0) \cong \mathbb{Z}$  (0 and 1 respectively). Since homotopic loops represent the same element of  $\pi_1$  this is a contradiction! Hence, no such homotopy  $f_t$  exists.